System—both in extent and in mass. They are the ships that first explored what may be homelands of our remote descendants.

U.S. launch vehicles are these days too feeble to get such a spacecraft to Jupiter and beyond in only a few years by rocket propulsion alone. But if we're clever (and lucky), there's something else we can do: We can (as *Galileo* also did, years later) fly close to one world, and have its gravity fling us on to the next. A gravity assist, it's called. It costs us almost nothing but ingenuity. It's something like grabbing hold of a post on a moving merry-go-round as it passes—to speed you up and fling you in some new direction. The spacecraft's acceleration is compensated for by a deceleration in the planet's orbital motion around the Sun. But because the planet is so massive compared to the spacecraft, it slows down hardly at all. Each *Voyager* spacecraft picked up a velocity boost of nearly 40,000 miles per hour from Jupiter's gravity. Jupiter in turn was slowed down in its motion around the Sun. By how much? Five billion years from now, when our Sun becomes a swollen red giant, Jupiter will be one millimeter short of where it would have been had *Voyager* not flown by it in the late twentieth century.

Voyager 2 took advantage of a rare lining-up of the planets: A close flyby of Jupiter accelerated it on to Saturn, Saturn to Uranus, Uranus to Neptune, and Neptune to the stars. But you can't do this anytime you like: The previous opportunity for such a game of celestial billiards presented itself during the presidency of Thomas Jefferson. We were then only at the horseback, canoe, and sailing ship stage of exploration. (Steamboats were the transforming new technology just around the corner.)

Since adequate funds were unavailable, NASA's Jet Propulsion Laboratory (JPL) could afford to build spacecraft that would work reliably only as far as Saturn. Beyond that, all bets were off. However, because of the brilliance of the engineering design—and the fact that the JPL engineers who radioed instructions up to the spacecraft got smarter faster than the spacecraft got stupid—both spacecraft went on to explore Uranus and Neptune. These days they are broadcasting back discoveries from beyond the most distant known planet of the Sun.